# Inventory System

Overview

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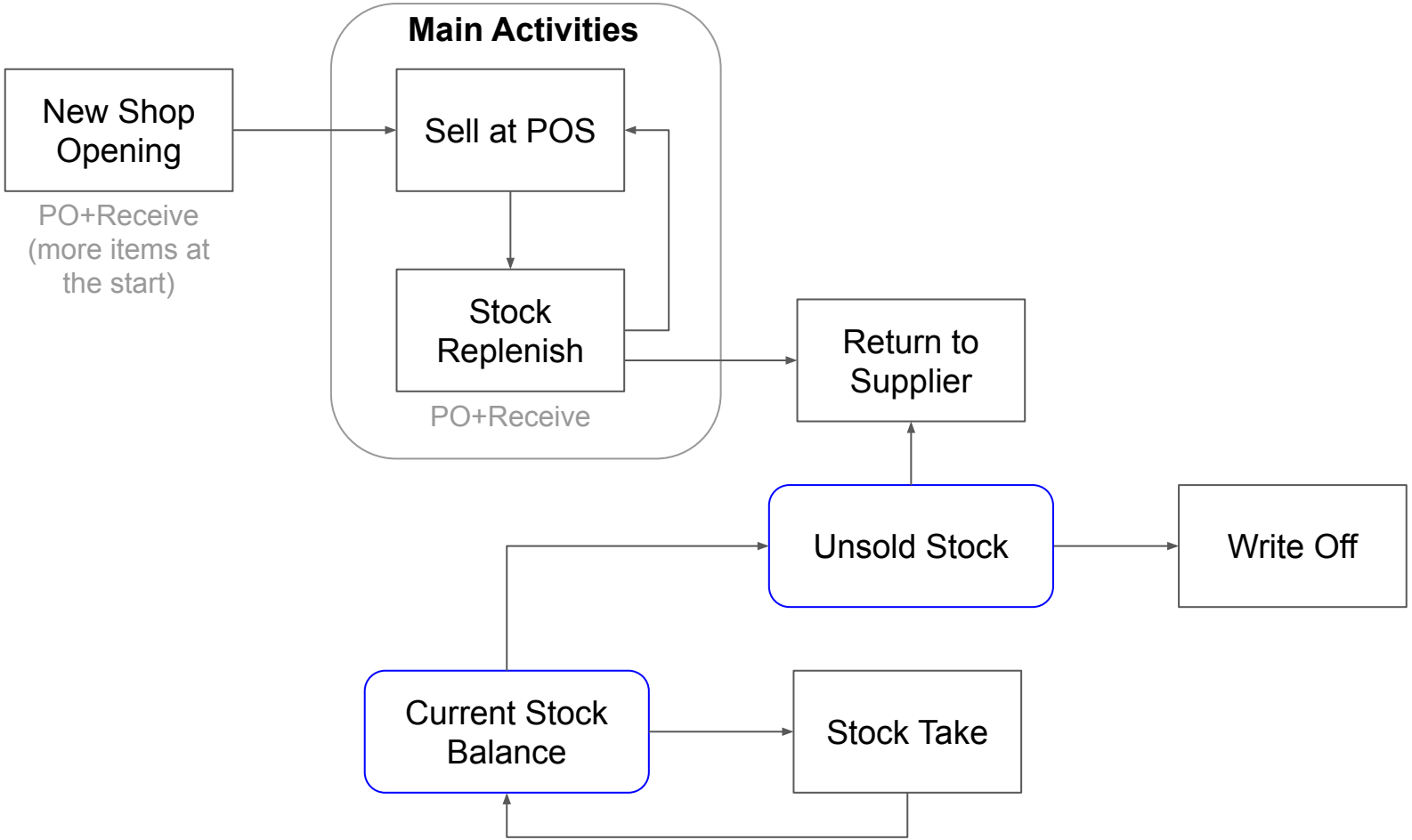
# Common Inventory Transactions


Main Activities
New Shop Opening
PO+Receive (more items at the start)
Sell at POS
Stock Replenish
PO+Receive
Return to Supplier
Unsold Stock
Write Off
Current Stock Balance
Stock Take

**Common Inventory Transactions**

1. New Shop Opening - loading new stock
   - lots of stock to receive for convenience stores

2. Send Purchase Order (PO) to supplier - order products from supplier

3. Receive Goods based on PO - stock in

4. Return Goods to supplier - stock return

5. Sales - deduct stock after sales
   - capture deducted stock as cost of sales or Cost of Goods Sold (COGS)

6. Write Off - stock out with reason

7. Stock Take - count actual stock and compare with system and apply the actual stock count to system
   - usually once a year for Retail, and once a month for F&B

# Common Inventory Reports

**Common Inventory Reports**

1. Stock Activity report - check stock activity details

2. Stock On Hand report - check current stock balance

3. Stock Valuation report - check current stock valuation

4. Stock Valuation at certain Date - get stock valuation at certain date (e.g.: year end) for Accounting

5. Stock Card Report - show all stock movements between 2 dates

# Average Cost

## Average Cost

1. When receiving new stock, calculate the average stock value by adding new stock value and old stock value divided by the total quantity of new + old stock.

2. During sales, after deducting the stock, the total average cost of deducted stock is recorded as Cost of Goods Sold (COGS) for each sales transaction

Old stock
$0.50 each
Value = $2.50

New stock
$0.60 each
Value = $3.00

Old + New stock
Total = $5.50 / 10
average cost
$0.55 each

Sell 2 pcs, COGS = 2 x $0.55 = $1.10

# FIFO Cost

**FIFO Cost**
(First In First Out)

1. When receiving new stock, record each stock at actual cost

2. During sales, after deducting the stock, the total actual cost of deducted stock is recorded as Cost of Goods Sold (COGS) for each sales transaction

We cannot support this costing method, because it's too tedious to track and look for every single item cost

Old stock
$500.00 each

New stock
$550.00 each

Sell 1st pc, COGS = $500.00
Sell 2nd pc, COGS = $500.00
Sell 3rd pc, COGS = $500.00

Sell 4th pc, COGS = $550.00
Sell 5th pc, COGS = $550.00

# Stock Replenishment

**Stock Replenishment**

1. Based on Average Sales History
   - system can auto recommend what and how much to order

2. Ad-hoc based on foreseeable demand, e.g.: holiday season, current trend, special event, etc
   - based on staff experience and judgement
   - staff can overwrite system recommendation

3. Auto based on demand forecasting
   - we cannot do this yet

# 1. Based on Average Sales History

| Item No | Item Name | UOM | Min | Max | Bal | PO | Wk-1 Sales | Wk-2 Sales | Wk-3 Sales | Avg Sales |
|---|---|---|---|---|---|---|---|---|---|---|
| 8888196173324 | POKKA JASMINE GREEN TEA 0.5L | 0.5L | 47 | 94 | 123 | 0 | 53 | 38 | 50 | 47 |
| 8888196170217 | POKKA LEMON TEA 1.5L | 1.5L | 21 | 43 | 28 | 0 | 29 | 23 | 12 | 21.33 |
| 8888196170019 | POKKA OOLONG TEA 1.5L | 1.5L | 15 | 31 | 57 | 0 | 14 | 19 | 13 | 15.33 |
| 8888196188915 | POKKA PEPPERMINT JASMINE GREEN TEA 0.5L | 0.5L | 15 | 29 | 84 | 0 | 12 | 11 | 21 | 14.67 |
| 9002490218058 | RED BULL ENERGY DRINK 355ML | 250ML | 14 | 27 | 1 | 0 | 3 | 21 | 17 | 13.67 |
| 8888196451767 | POKKA JASMINE GREEN TEA 6SX250ML | 6SX250ML | 10 | 20 | 10 | 0 | 6 | 12 | 12 | 10 |
| 8888196189011 | POKKA PEACH OOLONG TEA 0.5L | 0.5L | 10 | 19 | 5 | 0 | 5 | 11 | 13 | 9.67 |
| 8888196190314 | POKKA SOURSOP JUICE DRINK 0.5L | 0.5L | 9 | 18 | 34 | 0 | 9 | 10 | 8 | 9 |
| 8888196905819 | POKKA PREMIUM MILK TEA (LESS SUGAR) 0.5L | 0.5L | 9 | 17 | 16 | 0 | 6 | 2 | 18 | 8.67 |
| 8888196904515 | POKKA PREMIUM MILK COFFEE (LESS SUGAR) 0.5L | 0.5L | 8 | 17 | 5 | 0 | 7 | 9 | 9 | 8.33 |

Min = 1 week average sales
Max = 2 weeks average sales

Click here for Retail Purchase Order user guide

Stock balance is now 1 (drops to below min 14)
- system will recommend to buy 27 - 1 = 26

If minimum order qty is 6
- system will recommend to buy 30 (which is 5 multiples of 6)
- this will cause new stock to be 31 (above 27) after the order is placed

Before the order is received, the order qty will be tracked in PO qty
- Next order will not recommend to order again since the order has been placed
- after the PO is received, the PO qty will become Balance qty

# Goods Receive Discrepancies

a) If the reason is recoverable, e.g.: Transport sent to wrong outlet, Partial shipment (balance to ship later)
- Outlet A will receive the balance 2 pcs when Transport sends the balance

b) If the reason is not recoverable, e.g.: damaged or lost.
- Supplier will bear the cost
- Outlet A can close the partially received PO
- the final payment to supplier will be based on the actual received qty

c) If the reason is due to no stock
- Outlet A will close the partially received PO
- the final payment to supplier will be based on the actual received qty

# Improve
# Gross Profit Margin

# Gross Profit Margin = Revenue - COGS

## Product Sales Profit Report

Export

| TOTAL NETT SALES | TOTAL COST | TOTAL PROFIT | MARGIN% |
|---|---|---|---|
| $124,513.59 | $101,307.62 | $15,063.45 | 12.94% |

| Item No | Item Name | Quantity | Gross Sales | Discount | Nett Sales | GST | Cost | Cost% | Profit | Profit% | Margin% |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 021200510052 | 3M SCOTCHBRITE MULTIPURPOSE GLOVES MEDIUM 6PR/BG 24PR/CTN | 1.00 | 3.30 | 0.10 | 3.20 | 0.21 | 2.31 | 0.00% | 0.68 | 0.00% | 22.76% |
| 021200510069 | 3M SCOTCHBRITE MULTIPURPOSE GLOVES LARGE 6PR/BG 24PR/CTN | 2.00 | 6.60 | 0.10 | 6.50 | 0.43 | 4.62 | 0.00% | 1.45 | 0.00% | 23.94% |
| 029173000549 | PRUNE SLICED | 5.00 | 12.75 | 0.08 | 12.67 | 0.83 | 9.50 | 0.00% | 2.34 | 0.01% | 19.77% |
| 029173003397 | MARSHMALLOW - BBQ | 3.00 | 7.20 | 0.07 | 7.13 | 0.47 | 5.25 | 0.00% | 1.41 | 0.00% | 21.21% |
| 029173005230 | HONEY DATES | 7.00 | 14.70 | 0.12 | 14.58 | 0.95 | 11.20 | 0.01% | 2.43 | 0.01% | 17.80% |
| 029173006091 | STICK BISCUITS - TOMATO | 5.00 | 11.50 | 0.14 | 11.36 | 0.74 | 8.75 | 0.00% | 1.87 | 0.01% | 17.58% |
| 029173060543 | SWEET PRUNE | 9.00 | 17.55 | 0.00 | 17.55 | 1.15 | 13.50 | 0.01% | 2.90 | 0.01% | 17.69% |
| 029173073987 | CUTTLEFISH BBQ | 8.00 | 28.80 | 0.11 | 28.69 | 1.88 | 21.60 | 0.02% | 5.21 | 0.03% | 19.44% |
| 029173077626 | JELLY SWEET - L | 5.00 | 14.75 | 0.27 | 14.48 | 0.95 | 11.00 | 0.01% | 2.53 | 0.01% | 18.71% |
| 029173091332 | GARLIC GREEN PEAS | 6.00 | 18.00 | 0.09 | 17.91 | 1.17 | 13.50 | 0.01% | 3.24 | 0.02% | 19.34% |

Review low margin products

Strike a balance between sales volume and profit margin
Typically,
- high volume less margin
- low volume high margin

# Avoid Under-stocked to increase Sales

Sales drop due to low or no stock (customer cannot find the product to buy)

| Item No | Item Name | UOM | Min | Max | Bal | PO | Wk-1 Sales | Wk-2 Sales | Wk-3 Sales | Avg Sales |
|---|---|---|---|---|---|---|---|---|---|---|
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| 9002490218058 | RED BULL ENERGY DRINK 355ML | 250ML | 14 | 27 | 1 | 0 | 3 | 21 | 17 | 13.67 |
| 8888196451767 | POKKA JASMINE GREEN TEA 6SX250ML | 6SX250ML | 10 | 20 | 10 | 0 | 6 | 12 | 12 | 10 |

1. Last week sales drop from 17-21 to only 3 due to low stock item
2. Weekly sales is stable when stock is enough for 3 weeks average sales

# Avoid Over-stocked to improve Cash on hand

# Cash is tied up to stock that don't sell well (instead of being used to generate new profit)

| Item No | Item Name | UOM | Min | Max | Bal | PO | Wk-1 Sales | Wk-2 Sales | Wk-3 Sales | Avg Sales |
|---|---|---|---|---|---|---|---|---|---|---|
| 8888196188113 | POKKA KIYO KYOHO GRAPE JUICE 500ML | 500ML | 8 | 16 | 36 | 0 | 5 | 14 | 5 | 8 |
| 8888196904812 | POKKA WANG KOPI 0.5L | 0.5L | 8 | 15 | 4 | 0 | 12 | 6 | 5 | 7.67 |
| 8888196172211 | POKKA PEACH TEA 0.5L | 0.5L | 8 | 15 | 82 | 0 | 3 | 0 | 20 | 7.67 |
| 8888196903211 | POKKA PREMIUM MOCHA COFFEE 0.5L | 0.5L | 7 | 15 | 13 | 0 | 10 | 8 | 4 | 7.33 |
| 8888196902610 | POKKA JASMINE GREEN TEA (NO SUGAR) 0.5L | 0.5L | 7 | 15 | 40 | 0 | 12 | 7 | 3 | 7.33 |
| 8888196951410 | POKKA JASMINE GREEN TEA (NO SUGAR) 1.5L | 1.5L | 7 | 14 | 23 | 0 | 11 | 6 | 4 | 7 |

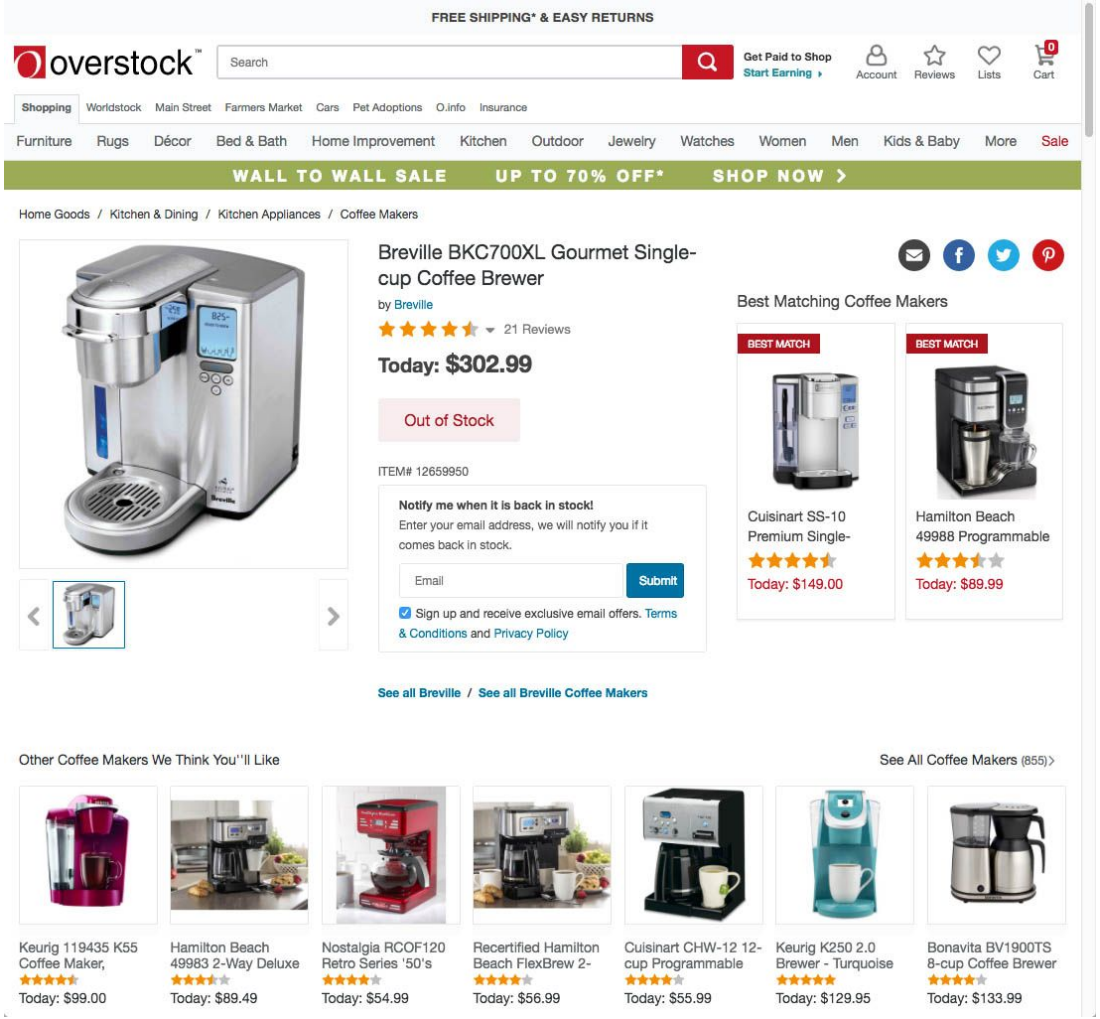
3 Possible overstock, stock balance > 4x average weekly sales

4 Possible overstock, stock balance > 10x average weekly sales

# Stock Level Availability for In-store customer service

Can you check which outlet has the stock ?

# Stock Level Availability for eCommerce




1. Ensure order placed by customer online can be delivered

2. Reduce Customer recovery needed for out of stock orders

# Stock Take

Stock Take

1. Stock Take on stock gun

2. Review Stock Take discrepancies (1st time)

3. Redo stock take on items with large or unusual discrepancies

4. Review stock take discrepancies (2nd time)

5. Redo stock take and review as many times as necessary typically 2x, max 3x

6. Approve Stock Take

7. Stock Take Qty will overwrite system qty
   - adjustment down will be recorded as wastage
   - adjustment up will be recorded as adjustment in

# Recipe / BOM

**Recipe**

1. Setup semi-finished products, like chilli sauce, nasi lemak rice, truffle sauce, etc.
2. Setup finished products to sell, like nasi lemak, truffle pasta, etc.

## Recipe Setup

Product Name

ALL

Material Name

Chicken Meat Raw

SEARCH ADD NEW SHOW RECIPE WITH WRONG CONVERSION

| Item No | Product Name | Total Material | Total Material Cost | Retail Price | |
|---|---|---|---|---|---|
| I00100000936 | Chicken Minced | 1 | $ 0.5000 | $ 1.00 | edit |
| I00100000935 | Chicken Sausage | 1 | $ 0.5000 | $ 1.00 | edit |

# Production / Item Cooking

## Production (Retail)

1. Produce new product and consume ingredients or raw material products

2. May have wastage (only in Retail)
   (F&B cannot handle wastage yet)

1 gallon = 5000ml

Can produce 20 bottles of 250ml

Produce 20 bottles of Hand Sanitizer, material used:

| Used Qty | Material | Total Cost |
|---|---|---|
| 5000ml | Hand Sanitizer | $20.00 |
| 20 bottle | Empty Bottle | $4.00 |
| 21 pc (1pc torn) | Label | $6.30 |
| **TOTAL COST** | | $30.30 |

- Stock Out all materials used (total cost = $30.30)
- Stock In 20 bottles of Hand Sanitizer at cost of $30.30 / 20 = $1.515 each

Bill of Material for Hand Sanitizer

| Req Qty | Material | Unit Cost |
|---|---|---|
| 250ml | Hand Sanitizer | $1.00 |
| 1 bottle | Empty Bottle | $0.20 |
| 1 pc | Label | $0.30 |
| TOTAL COST | | $1.50 |

**Item Cooking (F&B)**

1. Produce new product and consume ingredients or raw material products

2. May have wastage
   (F&B cannot handle wastage yet)

Produce 20,
material used:

| Used Qty | Material | Total Cost |
|---|---|---|
| 5000ml | | $20.00 |
| 20 bottle | | $4.00 |
| 21 pc (1pc torn) | | $6.30 |
| **TOTAL COST** | | $30.30 |

- Stock Out all materials used (total cost = $30.30)
- Stock In 20 bottles of Hand Sanitizer at cost of $30.30 / 20 = $1.515 each

Recipe

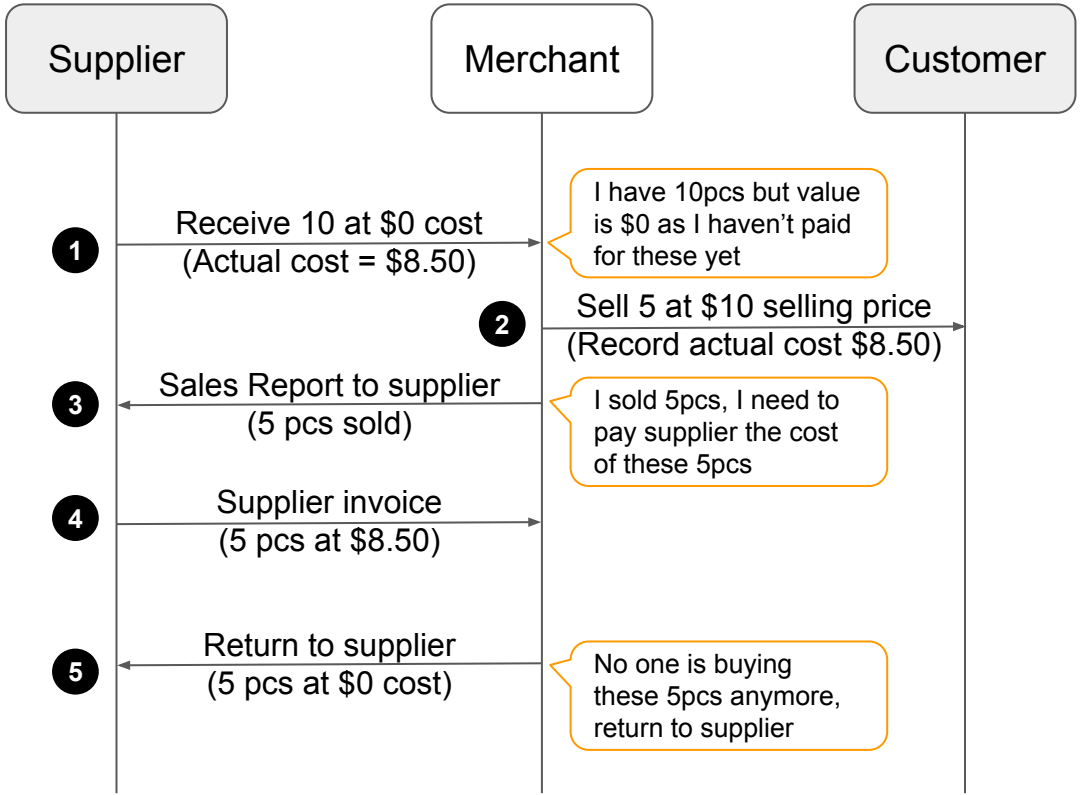
| Req Qty | Material | Unit Cost |
|---|---|---|
| | | $1.00 |
| | | $0.20 |
| | | $0.30 |
| TOTAL COST | | $1.50 |

# Pack Size

## Pack Size

1. Click the link for Retail Pack Size guide

# Consignment from Supplier



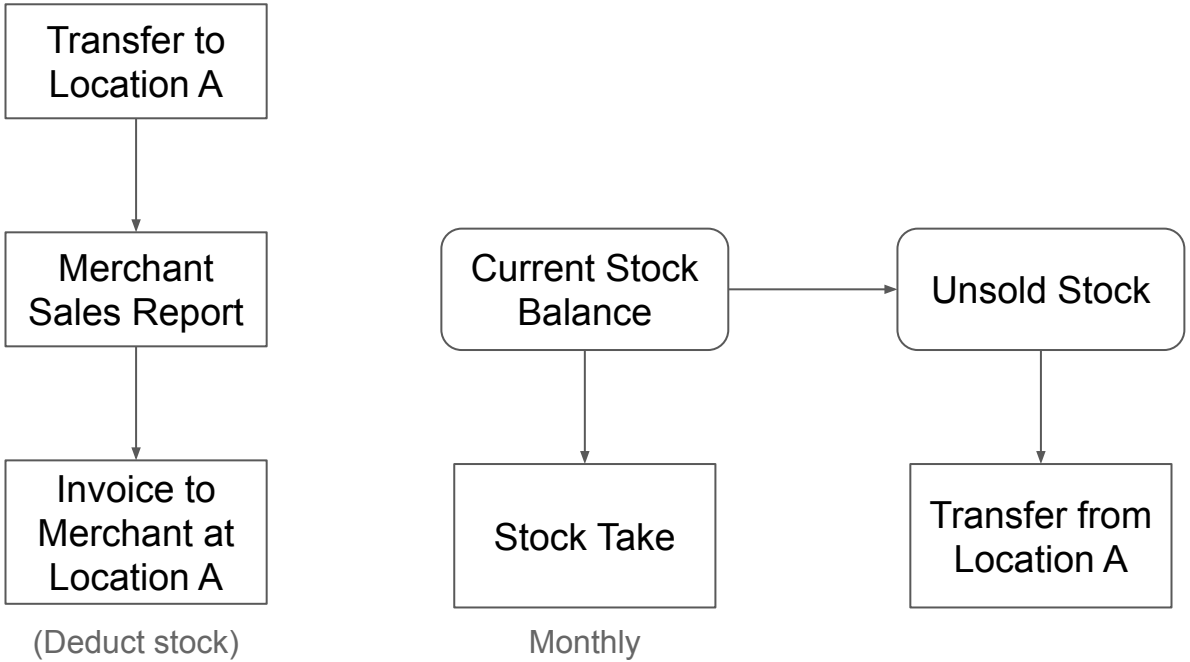

1. Consignment goods are received as stock by Merchant at $0 cost

2. Need to create supplier invoice only after the goods is sold to customer

3. Merchant can return unsold stock balance to supplier at $0 cost

This feature is not ready yet

# Consignment to Others




Consign products to other merchant to sell

E.g.: Department store, bigger supermarket, etc

# Multi Outlet with Warehouse

**Multi Outlet with Warehouse**

1. Order from Warehouse - Outlet requests stock from Warehouse

2. Return to Warehouse - Outlet returns stock to Warehouse

3. Stock Transfer - transfer stock from one outlet to another outlet

Click here to view the F&B Outlet Ordering user guide

# Outlet Order Discrepancies

If the reason is recoverable, e.g.: Transport sent to wrong outlet
- Outlet A will receive the balance 2 pcs when Transport sends the balance

Assuming that the root cause is not recoverable, e.g.: damaged or lost.

a) If Warehouse agreed to bear the shortage cost
   - Warehouse has to change the approved qty to 10 pcs, the order will be closed
   - Warehouse will perform Stock Out to write off the 2 pcs as damaged or lost

b) If Outlet A agreed to bear the shortage cost
   - Outlet A will receive the balance 2 pcs, the order will be closed
   - Outlet A will perform stock out to write off the 2 pcs as damaged or lost

# Stock Transfer Discrepancies

If the reason is recoverable, e.g.: Transport sent to wrong outlet
- Outlet B will receive the balance 2 pcs when Transport sends the balance

Assuming that the root cause is not recoverable, e.g.: damaged or lost.

a) If Outlet A agreed to bear the shortage cost
   - Outlet A has to change the transfer qty to 10 pcs, the transfer will be closed
   - Outlet A will perform Stock Out to write off the 2 pcs as damaged or lost

b) If Outlet B agreed to bear the shortage cost
   - Outlet B will receive the balance 2 pcs, the order will be closed
   - Outlet B will perform stock out to write off the 2 pcs as damaged or lost